Ex. No: 10

Record-IMPLEMENTATION OF OPENING AND CLOSING

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In [41]:
```
import cv2
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
```

In [42]:
```
def load_img():
    blank_img = np.zeros((600,600),dtype=np.uint8)
    font = cv2.FONT_HERSHEY_SIMPLEX

    cv2.putText(
        blank_img,
        text="Dharsh",
        org=(50,300),
        fontFace=font,
        fontScale=5,
        color=(255,255,255),
        thickness=25,
        lineType=cv2.LINE_AA
    )

    return blank_img
```

OPENING

In [22]:
```
img = load_img()
```

In [23]:
```
white_noise = np.random.randint(low=0,high=2,size=(600,600))
```

In [24]:
```
white_noise
```

Out[24]:
```
array([[1, 1, 0, ..., 1, 0, 1],
       [1, 1, 0, ..., 0, 1, 1],
       [0, 0, 0, ..., 0, 1, 0],
       ...,
       [1, 0, 1, ..., 1, 1, 0],
       [0, 1, 1, ..., 1, 0, 1],
       [0, 0, 1, ..., 1, 1, 1]], shape=(600, 600), dtype=int32)
```

In [25]:
```
img.max()
```

Out[25]:
```
np.uint8(255)
```

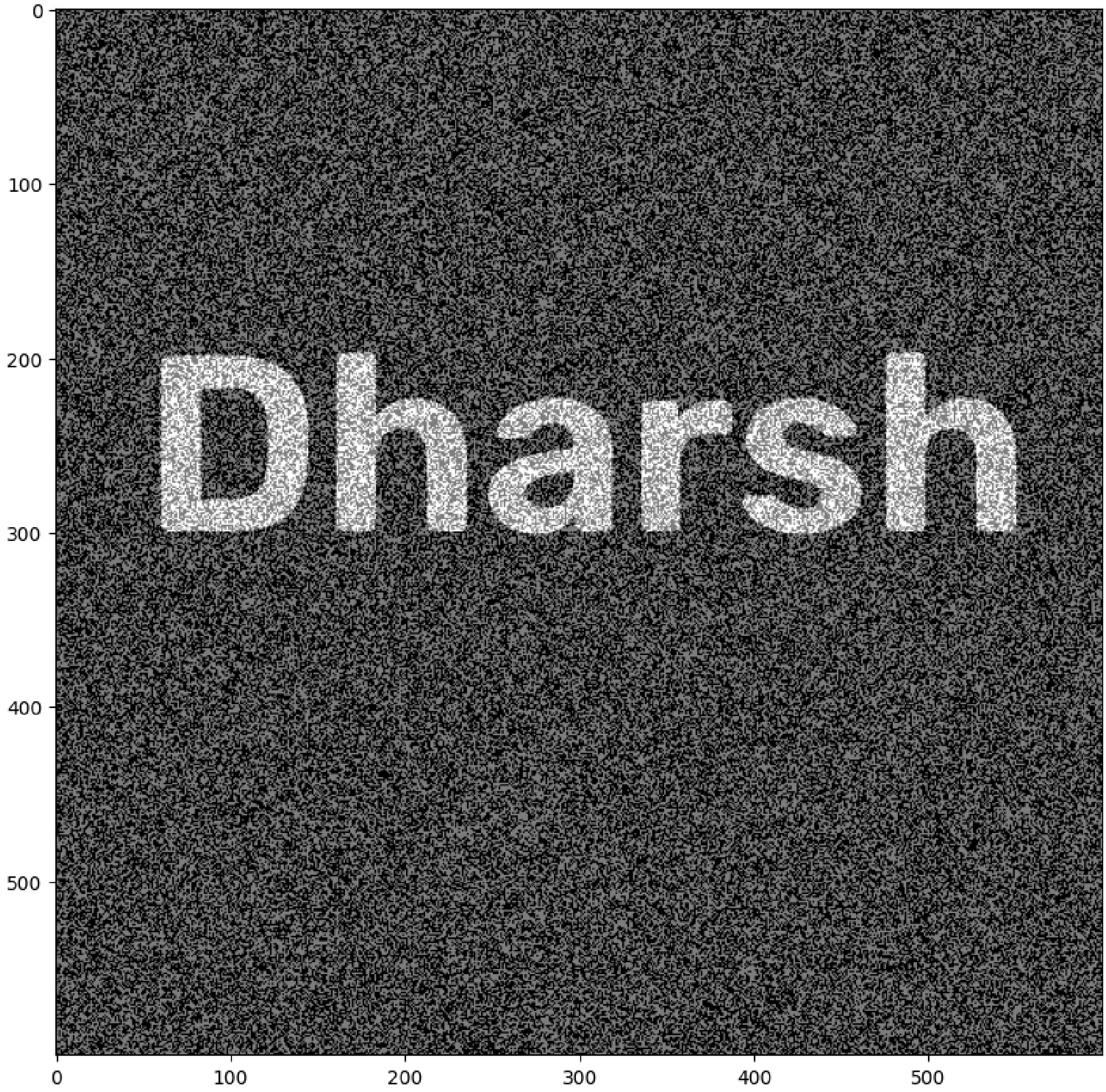
In [26]:
```
white_noise = white_noise*255
```

In [27]:
```
white_noise
```

Out[27]:
```
array([[255, 255,   0, ..., 255,   0, 255],
       [255, 255,   0, ...,   0, 255, 255],
       [  0,   0,   0, ...,   0, 255,   0],
       ...,
       [255,   0, 255, ..., 255, 255,   0],
       [  0, 255, 255, ..., 255,   0, 255],
       [  0,   0, 255, ..., 255, 255, 255]], shape=(600, 600), dtype=int32)
```

In [28]:
```
noise_img = white_noise+img
```

In [29]:
```
display_img(noise_img)
```



In [31]:
```
opening = cv2.morphologyEx(img, cv2.MORPH_OPEN, kernel)
display_img(opening)
```

CLOSING

In [32]:
```
img = load_img()
```

In [33]:
```
black_noise = np.random.randint(low=0,high=2,size=(600,600))
```

In [34]:
```
black_noise
```

Out[34]:
```
array([[0, 1, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 1, 0, 1],
       [1, 1, 1, ..., 1, 0, 1],
       ...,
       [1, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 1, 1, 1],
       [1, 0, 1, ..., 1, 0, 0]], shape=(600, 600), dtype=int32)
```

In [35]:
```
black_noise= black_noise * -255
```

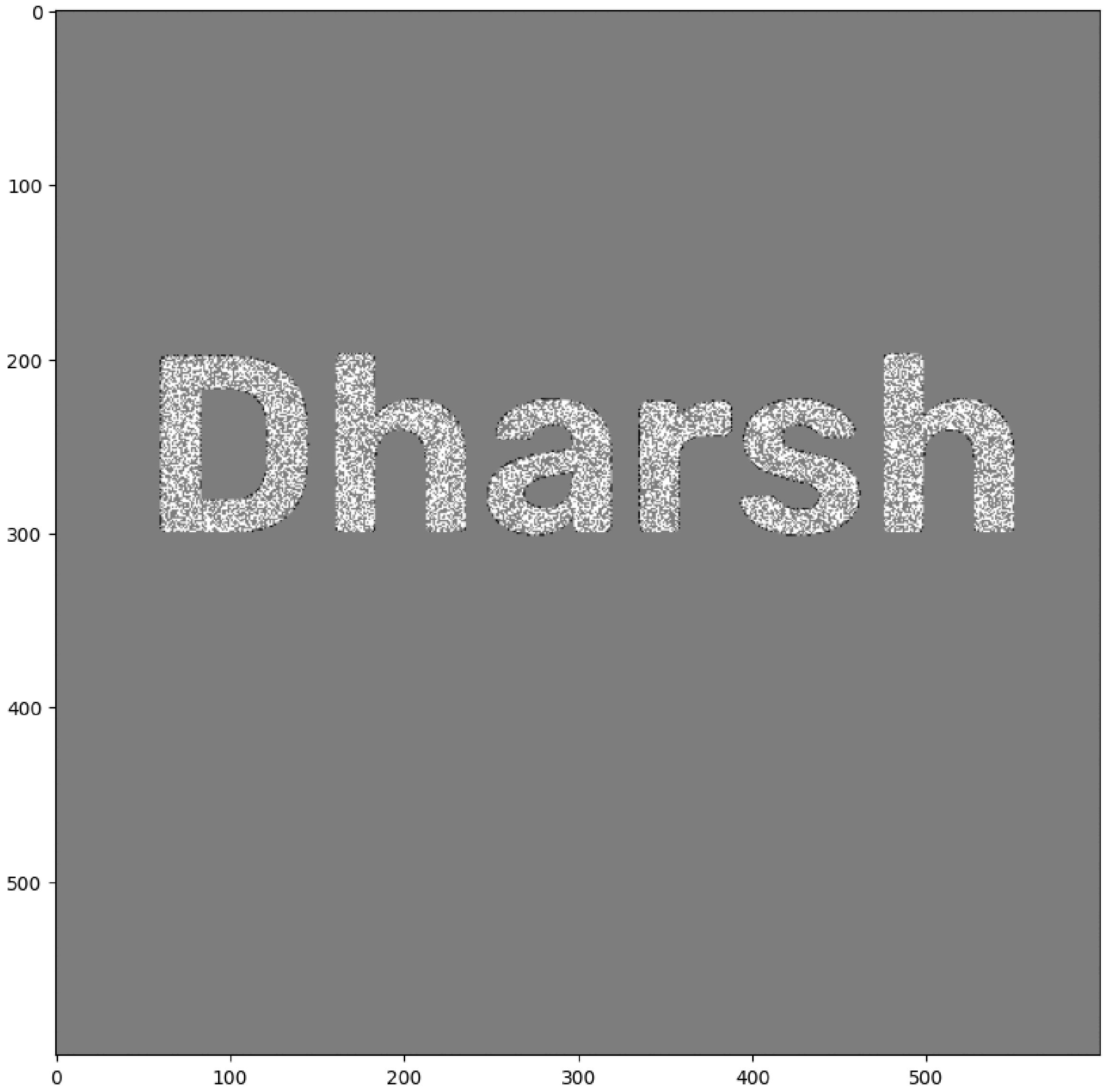
In [36]:
```
black_noise_img = img + black_noise
```

In [37]:
```
black_noise_img
```

Out[37]:
```
array([[   0, -255,    0, ...,    0,    0,    0],
       [   0,    0,    0, ..., -255,    0, -255],
       [-255, -255, -255, ..., -255,    0, -255],
       ...,
       [-255,    0,    0, ...,    0,    0,    0],
       [   0,    0,    0, ..., -255, -255, -255],
       [-255,    0, -255, ..., -255,    0,    0]],
      shape=(600, 600), dtype=int32)
```

In [38]:
```python
black_noise_img[black_noise_img==-255] = 0
```

In [39]:
```python
display_img(black_noise_img)
```



In [40]:
```python
closing = cv2.morphologyEx(img, cv2.MORPH_CLOSE, kernel)
display_img(closing)
```

In [ ]: